Model Analysis Report

Storm Sequence Generator

MKM-SS-001

Report Date: 27 March 2026

Governance Framework: SR 11-7 / SS1/23 Model Risk Management

Contents

1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Expected storm events per year by mean inter-storm gap.

Mean gap (days)	Expected events/yr	Simulated (seed=42)
3	121.7	131
5	73.0	84
7	52.1	58
10	36.5	28
14	26.1	24
21	17.4	19
30	12.2	11

Table 2: Aggregate severity by storm sequence length (base 50 mm/hr, decay 0.85).

Storms in cluster	Peak (mm/hr)	Total (mm/hr)	Mean (mm/hr)
1	50.0	50.0	50.0
2	50.0	92.5	46.2
3	50.0	128.6	42.9
5	50.0	185.4	37.1
7	50.0	226.5	32.4
10	50.0	267.7	26.8

Table 3: Cluster total variance by intensity correlation ($n = 3$, $\sigma = 15$).

Correlation ρ	Cluster variance	Cluster std dev
0.0000	675.0	26.0
0.2000	945.0	30.7
0.4000	1,215.0	34.9
0.6000	1,485.0	38.5
0.8000	1,755.0	41.9
1.00	2,025.0	45.0

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 4: Stress testing scenarios for Storm Sequence Generator.

Scenario	**Parameter Shock**	**Severity**	**Status**
Baseline	No shock	—	Passed
Moderate stress	$+2\sigma$ inputs	Medium	Pending
Severe stress	$+3\sigma$ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 5: Performance metrics for Storm Sequence Generator.

Metric	**Value**	**Status**
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.